



Remote · Hybrid · Spain

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SKILLS

LLM & AGENTS

MCPA2ARAG/GraphRAG

LangChainLangGraph

Google ADKLiteLLMLangfuse

Neo4jHuggingFaceQdrant

MLOPS & INFRA

DockerKubernetesHelm

TerraformMLflowCI/CD

FastAPIOpenTelemetryRedis

DATA & STREAMING

KafkaApache Flink

ElasticsearchPostgresPySpark

DatabricksSQL

CLOUD

GCP Vertex AIGeminiFirebase

AWSAzure ML

DEV STACK

PythonNext.jsReact Native

Expo/SolitoTurborepoNx

Git

LANGUAGES



EDUCATION

- MSc Artificial Intelligence
- B.Sc. Systems Engineering
- Lic. en Psicología
- Best Graduate

CLINICAL EXPERTISE

Licensed Psychologist (5+ years). Diagnostic precision applied to LLM evaluation (HITL), hallucination detection, and AI safety design for mental-health systems.

Martin Rodrigo Morales

AI Technical Lead · MLOps Engineer · AI Systems Architect

MSc Artificial Intelligence · Lic. en Psicología · B.Sc. Systems Engineering

SUMMARY

AI Technical Lead & MLOps Engineer with 5+ years building production-grade LLM and agentic systems. Currently at **latel** leading a 4-person team on **+OAA / NOA**, an event-driven multi-agent platform running in production for a leading Spanish telecom operator, automating **root-cause** diagnosis of network failures. Co-founder and technical coordinator of **Dualtherapist** (clinical AI), approaching launch. **Architect** profile: agentic systems (MCP, A2A, BPM), RAG/GraphRAG, **LLMOps**, and end-to-end cloud architecture. Proven track record in **Fortune 500** environments (Accenture, BBVA/HP) and a unique position at the intersection of Systems Engineering and Clinical Psychology.

EXPERIENCE

- AI Technical Lead
- latel · Telecommunications / Enterprise AI · Spain
- Leading a 4-person team building **+OAA / NOA**, a production multi-agent platform that automates root-cause detection of network failures for a leading Spanish telecom operator.
- Event-driven microservices architecture: multi-channel ingestion (syslog, mail, SMS, Telegram) → **Kafka (Avro) + Apache Flink** → Elasticsearch, with agent orchestration via **Flowable BPM (BPMN)** and an agent HTTP bridge.
- Designed the in-house **LLMOps** layer: **LiteLLM + Langfuse** gateway (governance, virtual keys, cost control and observability) and a self-hosted **Qwen3-30B model on GKE**, with a RAG vs. fine-tuning/distillation (RAFT) strategy.
- LLM reasoning pipeline over live network telemetry for root-cause classification and an **agentic chat with MCP** and multi-step reasoning. Stack: Next.js/React 19 (Nx), FastAPI + Postgres, OpenTelemetry, Docker, Kubernetes + Helm.

- Independent AI Consultant & Professor
- Northern International University / Self-employed
- Appointed professor at NIU delivering modules on AI-Assisted Psychometrics, Computational Psychopathology, and Conversational Therapeutic Systems.
- Built scalable RAG pipelines for the health and education sectors integrating HuggingFace LLMs with Qdrant/ChromaDB vector stores.
- Mentored 20+ students through the full NLP project lifecycle: LangChain architecture, FastAPI deployment, and production best practices.

- AI Engineer
- Accenture · Client: HP (Fortune 500)
- Designed and deployed a RAG-based candidate evaluation system (MinHash near-dedup, perplexity filtering, domain rebalancing), improving **NDCG@5 0.71→0.83**.
- Led MLOps strategy for HP supply-chain predictive models: PySpark pipelines, Docker + Databricks CI/CD.
- Optimized Kubernetes cluster resource allocation and autoscaling, reducing cloud infrastructure costs.

- Lead Data Instructor & Technical Mentor
- Henry Bootcamp
- Led 10+ cohorts as Technical Lead & Scrum Master through 45-day production-simulation sprints.
- Co-designed cloud-native architectures (Docker, FastAPI, MLflow) deployed by graduates to real production environments.

- Big Data Engineer
- Accenture · Client: BBVA (Banking)
- Orchestrated mission-critical ETL pipelines in a regulated banking environment using Hadoop, Spark on YARN, and GCP Dataproc.
- Fixed dataloader starvation (repartition to 256 MB files + speculative execution), improving throughput **~2.5x**; Data Quality & Drift monitoring with Control-M.

RESEARCH

APOLA

Structural Psychoanalysis — International Forums.

Topology

Set Theory, Logic & Language formalization.

KG Theory

Topological structures informing Knowledge Graph design.

DOMAIN FOCUS

- Mental Health AI
- Clinical NLP
- AI Alignment
- HITL
- HealthTech
- FinTech
- Telco / Networks

LEADERSHIP

- Technical Lead & Scrum Master
- Cross-domain thinking: Tech × Clinical
- Mentorship & knowledge transfer



ACADEMIC EXCELLENCE

Gold Medal — Best Graduate
Universidad de Mendoza, 2017

SELECTED PROJECTS

Dualtherapist — Clinical AI Platform

CO-FOUNDER

Turbo · Next.js 15 · Expo/Solito · FastAPI · Gemini/Vertex AI · Neo4j · Firestore · Docker/K8s/Terraform

- Coordinating a 4-person team (frontend, backend, PM) building a digital therapy platform connecting psychologists and patients, approaching launch; responsible for technical architecture and agent system design.
- End-to-end architecture: **Next.js 15** web (App Router/SSR) + **Expo** mobile app with universal components (Solito), a Node/Express API, and a Python/**FastAPI** **ai-service** for session transcription, **embedding-similarity** triage, and a **RAG** knowledge base powered by Gemini/Vertex AI.
- Clinical knowledge graph in **Neo4j**; Firestore + Google Cloud Storage; Firebase Auth + JWT HttpOnly cookie with **Redis** blacklist; Docker / Kubernetes / Terraform infrastructure.
- Designed the **HITL** evaluation and the domain-specific clinical safety layer, drawing on my training as a licensed psychologist.

Multi-Agent Knowledge Graph Builder

Google ADK 1.5 · LiteLLM · Neo4j · Python 3.11

- Orchestrated autonomous agents for entity extraction and Neo4j knowledge-graph construction from unstructured data.
- Two-stage chunking (coarse semantic + fine structural) improved entity extraction **+8 F1**; full pipeline from document ingestion to graph queries.

MetaAgent — Hybrid Memory AI System

LangChain · FAISS + Qdrant · OpenAI API · Docker

- Persistent AI assistant with hybrid vector memory (FAISS + Qdrant), human-feedback scoring loops, and long-term context retention.
- Containerized with Docker; session persistence with automated memory consolidation and a scalable deployment pipeline.

Production MLOps Pipeline — Sentiment Analysis

DistilBERT · FastAPI · MLflow · HuggingFace Spaces · CI/CD

- Deployed a production DistilBERT model (80% accuracy) on HuggingFace Spaces with a Gradio UI and automated CI/CD for model versioning.
- Training optimization (fp16 + gradient checkpointing, **38→22 GB**) and INT8 quantization with ONNX Runtime (**p99 280→95 ms**).

PREVIOUS EXPERIENCE

Junior Data Scientist

2021 — 2023

CEA Solidario (NGO)

- Built AI-driven behavioral analysis tools for autism assistance, integrating early ML models for pattern recognition and behavioral insights.

CONTINUOUS DEVELOPMENT

- AI techniques: Prompt Engineering, Fine-tuning, RAG, Transfer Learning — HuggingFace Agents Course.
- React Native (Expo), advanced backend scalability, AWS serverless architecture (VPC, IaC).
- Specialization in Cognitive Neuroscience — Neuro-Transmitiendo (2024).